Employee response accuracy on repeated questions: forgetting and persistent errors in a three-year microlearning program

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Abstract

Background. Spaced repetition improves retention, but the effectiveness of microlearning in organizations has rarely been assessed using multi-year data on employee responses. **Methods.** The analysis covered 57,997 responses from 180 bank employees to 164 multiple-choice questions, given from May 2023 to June 2026. Questions recurred on a schedule independent of response outcome (median interval 37 days), and the correct answer with an explanation was displayed after every response. Accuracy was analyzed across successive attempts at the same question and by the outcome of, and interval since, the previous attempt. **Results.** Accuracy at the first attempt was 82.6%. In sequences of at least 5 attempts, it rose from 81.9% to 89.0% (by 7.2 pp, 95% CI [5.5, 8.9]). Error corrections outnumbered forgetting events (3382 vs. 2719). The forgetting rate increased with the interval and decreased with the number of consecutive correct responses, from 11.9% after one to 3.2% after at least 4. In 71.8% of cases, repeated errors involved the same distractor (expected under choice proportional to distractor popularity: 60.0%), regardless of the interval. **Conclusions.** Accuracy on recurring questions increased, but knowledge required further repetition, and some errors persisted despite feedback. A schedule adapted to employee performance and feedback addressing the selected distractor may increase the effectiveness of such programs.

Keywords: spaced repetition; microlearning; employee training; feedback; forgetting; learning analytics

1 Introduction

Retrieving information from memory strengthens it more than rereading does (the testing effect), and learning distributed over time is more durable than massed learning (the spacing effect) [4, 13]. Both effects are combined in spaced education, in which short multiple-choice questions recur periodically over weeks or months. In studies involving physicians and other health professionals, it improved the acquisition and retention of knowledge [7, 10].

In organizations, a similar format is used in *microlearning*, that is, short training units, often in the form of quizzes. Evidence of its effectiveness, however, comes mainly from studies with small samples and from grey literature [16], and the effects of mandatory employee training, e.g., in information security, tend to be small or to fade over time [12]. Training effectiveness is usually

assessed through transfer, that is, the application of knowledge at work [1]. Transfer, however, requires that the knowledge first be acquired and retained.

This earlier stage can be examined using data from microlearning platforms. The platform records each response together with its time and outcome, and the same question returns to the same employee repeatedly. Successive responses are therefore repeated measurements of the same knowledge in the same person. Similar data are used in learner models applied in intelligent tutoring systems and language learning applications [5, 14].

The study was conducted in a bank in which employees answered short questions displayed on their computers during the working day, and the system collected the responses over three years. Questions recurred on a schedule independent of response outcome, and the correct answer was displayed after every response. Such a schedule makes it possible to compare error corrections and forgetting without the selection introduced by adaptive algorithms, which is rare in data from training platforms. Laboratory studies further indicate that a selected distractor can become fixed in memory and that errors made with high confidence can recur despite feedback [2, 3]. Whether this phenomenon occurs in employee training is not known. Three research questions were formulated: (RQ1) does the percentage of correct responses increase when the same question returns to the same employee; (RQ2) how does accuracy depend on the outcome of the previous attempt and on the time elapsed since it; (RQ3) do repeated errors persist despite feedback, which would point to misconceptions rather than to guessing.

2 Materials and methods

2.1 Training program

The study was conducted in a bank. The questions were prepared by the *Banking Process Support and Optimization Department*, which also decided on employee participation in the program. The questions concerned knowledge used in everyday work at the bank. Each was a multiple-choice question with three to five answer options, one of which was correct, while the others were *distractors*. The questions, answer options and explanations were written in Polish.

The training application started automatically on working days on employees' computers and displayed one to four randomly selected questions per day, between 9:00 and 16:00. After each response, correct or incorrect, it showed the correct answer with an explanation (*feedback*). The same question could return to an employee no earlier than after seven days, and the repetition schedule did not depend on the response outcome. The program was suspended while the questions were reviewed and updated. In the analyzed data, the median number of responses per employee on an active day was 3 (IQR 2–4). The same question returned after a median of 37 days (IQR 18–90).

An example question presented to program participants, on cybersecurity, read: "Which of the following situations is phishing?". Answer options: "1 - A DDoS attack", "2 - Installing antivirus software", "3 - Backing up data", "4 - A fake e-mail with a link to the bank's login page (marked as the correct answer)". Explanation displayed after the response: "Phishing is a fraud method in which someone impersonates a trusted person or institution (e.g., a bank, a courier, a public office) to trick you into revealing confidential data such as passwords, card numbers or logins. Most often it looks like an e-mail from the 'bank' with an urgent request to log in, a text message with a link to an 'extra payment for a parcel', or a fake website deceptively similar to the real one. The key mechanism is that the fraudster creates a sense of urgency or stress so that you click without thinking. IMPORTANT: do not click suspicious links, always check the website address (URL), do not enter your login details via a link from a message, and if something looks 'too urgent', it is often a scam.".

2.2 Data and study population

The recorded employee responses were analyzed. Each record contained the question, the selected option, the employee code, the date and time of the response, and whether the response was correct. The analytical dataset comprised 57,997 responses (Table 1). The bank also provided a summary of employees' age and sex.

Table 1: Characteristics of the dataset.

Characteristic	Value
Responses	57,997
Employees	180
Questions	164
Observation period	2023-05-10 – 2026-06-08
Employee–question sequences	15,270
Observed answer options per question, median (range)	3 (2–5)
Responses per employee, median [IQR]	195 [70, 575]
Questions per employee, median [IQR]	86 [38, 138]
Active days per employee, median [IQR]	84 [30, 224]
Responses per active day, median [IQR]	3 [2, 4]
Months between first and last response, median [IQR]	18 [5, 30]
Attempts per sequence, median [IQR] (max.)	3 [2, 5] (17)
Sequences with $\geq$2 / $\geq$3 attempts (%)	78.1 / 59.8
Days between attempts, median [IQR]	37 [18, 90]
Correct responses, overall (%)	86.6
Correct responses, first attempts (%)	82.6

IQR: interquartile range. A sequence is a series of attempts by one employee at one question. The number of observed answer options may be smaller than the number of options in the question because some options were not chosen by any employee. Excluded test responses: 212. Excluded questions with a single observed answer option: 2 (348 responses). Excluded repetitions given earlier than 7 days after the previous attempt: 88.

Demographic data were available for 151 of the 180 employees (83.9%). Women accounted for 80.1% of persons of known sex ($n = 121$). The median age was 47 years (IQR 41–56; range 22–67), and the largest age group was 50–59 years (34.0%; Table 2). Age was included in the analysis of the first attempt and of the rate of accuracy gain; sex was not analyzed.

Table 2: Demographic characteristics of the employees included in the analysis.

Characteristic	Value
Employees included in the analysis	180
of whom with demographic data, n (%)	151 (83.9)
Women, n (%)	121 (80.1)
Men, n (%)	30 (19.9)
Sex not recorded, n	29
Age (years), median [IQR]	47 [41, 56]
Age (years), mean (SD)	47.2 (10.6)
Age (years), range	22–67
20–29 years, n (%)	13 (8.7)
30–39 years, n (%)	21 (14.0)
40–49 years, n (%)	49 (32.7)
50–59 years, n (%)	51 (34.0)
60–69 years, n (%)	16 (10.7)
Age not recorded, n	30

Percentages by sex were calculated among persons of known sex, and percentages by age group among persons of known age ($n = 150$). Missing data concern mainly employees absent from the demographic summary (29). IQR: interquartile range; SD: standard deviation.

In the response database, employees were identified by a random code, and the demographic summary was linked to the recorded responses using this code. Such a code cannot be reconstructed from a list of employees; the analyzed data were therefore fully anonymized.

2.3 Statistical analysis

The responses of one employee to one question, ordered in time, formed a *sequence*. The position of a response in the sequence was termed the *attempt number*, and the first response, that is, the employee's first encounter with the question, the *first attempt*. The *interval* was the number of days since the previous attempt in the same sequence, and *accuracy* was the percentage of correct responses. Accuracy was reported with 95% confidence intervals (CI) obtained with the bootstrap: employees were resampled with replacement (2000 replicates), which accounts for the dependence between responses given by the same person. In regression models, standard errors were clustered by employee.

First attempt. Accuracy at the first attempt was compared across age groups. The association with age (10-year groups) was estimated by logistic regression with question fixed effects, that is, with a separate intercept for each question. Question difficulty was not recorded in the data, so the fixed effects were estimated from the responses of all employees. The model thus compared employees of different ages answering the same questions. For comparison, it was also fitted without question fixed effects.

Changes within a sequence (RQ1). Accuracy was calculated for successive attempt numbers; attempts 10 and later were pooled. With each attempt, the number of sequences decreased, and employees who continued responding could differ from the others, which distorts aggregate curves [9]. The change was therefore also assessed in a *cohort*, that is, in sequences of at least 5 attempts, restricted to attempts 1 to 5. Accuracy at each of these attempts was calculated for the same sequences, and the measure of change was the difference between attempt 5 and attempt 1 in percentage points (pp). The association between the rate of gain and age was tested in the cohort by logistic regression with an interaction between $\log_2$ attempt number and age, and with employee

and question fixed effects.

Outcome of the previous attempt and interval (RQ2). Each attempt from the second onward was assigned to one of four categories according to the outcome of that attempt and of the previous one: a correct response after a correct one, *error correction* (correct after incorrect), *forgetting* (incorrect after correct) or *repeated error* (incorrect after incorrect). Because the error-correction rate (r) and the forgetting rate (f) have different denominators, the numbers of both events were compared, and the equilibrium level $r/(r + f)$, toward which accuracy would tend at constant rates, was determined. The rates were compared across six interval bins: 7–13, 14–29, 30–59, 60–119, 120–239 and at least 240 days. The forgetting rate was also tabulated by the number of consecutive correct responses before a given attempt and separately after an error correction. An incorrect response at the attempt following an error correction was termed an *error relapse*, and it was checked whether the employee chose the same distractor as before the correction. It was also checked whether the repetition schedule depended on the outcome of the previous attempt.

Persistent errors (RQ3). Accuracy was compared by the number of *consecutive errors*, that is, successive incorrect responses immediately preceding a given attempt, from 0 to at least 4. For two successive errors, the frequency with which the employee chose the same distractor was calculated and compared with the percentage expected under random choice and under choice proportional to distractor popularity in the given question. This percentage was also tabulated in three interval bins: shorter than 30 days, 30–89 days and at least 90 days. The share of repeated errors concentrated in the 10% of questions with the most such errors was also calculated.

Computations were performed in Python 3.14.4 using pandas 3.0.3, NumPy 2.5.0, SciPy 1.18.0, statsmodels 0.14.6 and Matplotlib 3.11.0.

3 Results

3.1 First attempt

At the first encounter with a question (15,270 responses from 180 employees), accuracy was 82.6% (95% CI [81.2, 84.1]; Table 3). Accuracy varied with age non-monotonically: it was lowest in the 30–39 age group (74.9%) and highest in the 50–59 age group (84.3%). When responses to the same questions were compared, the odds of a correct response increased with age (OR 1.16 per 10 years, 95% CI [1.03, 1.31]; $p = 0.012$). Without question fixed effects, the association was weaker (OR 1.12, 95% CI [1.01, 1.24]; $p = 0.039$).

Table 3: Accuracy at the first attempt, overall and by employee age group.

Age group (years)	Overall	20–29	30–39	40–49	50–59	60–69	Missing
Employees	180	13	21	49	51	16	30
First attempts, n	15,270	970	2011	4338	5096	1416	1439
Accuracy (%)	82.6	82.5	74.9	84.0	84.3	82.6	83.5
95% CI	[81.2, 84.1]	[77.5, 89.3]	[68.8, 80.7]	[81.5, 86.3]	[82.2, 86.5]	[78.7, 86.4]	[80.0, 86.8]

The first attempt is the first response in a sequence. 95% CIs were obtained with the bootstrap. The association with age was estimated by the logistic regression described in Section 2.3 (13,578 responses from 150 employees of known age): OR 1.16 per 10 years, 95% CI [1.03, 1.31], $p = 0.012$.

3.2 Changes within a sequence (RQ1)

Accuracy increased with attempt number, from 82.6% [81.1, 84.1] at the first attempt to 92.7% [90.6, 94.8] at attempt 10 and later (Figure 1). Because the number of sequences decreased with

each attempt, the change was examined in a cohort of 4890 sequences. In the cohort, accuracy rose from 81.9% [79.6, 84.1] at attempt 1 to 89.0% [87.3, 90.7] at attempt 5, that is, by 7.2 pp (95% CI [5.5, 8.9]; Table 4). An increase within the same sequences means that the improvement did not result from the earlier termination of sequences with lower accuracy. Accuracy at attempt 1 in the cohort (81.9%) was close to accuracy at the first attempt in all sequences (82.6%).

The estimated increase was positive in all age groups and ranged from 4.0 pp (60–69 years) to 8.8 pp (40–49 years), although in the small extreme age groups the lower limit of the 95% CI was close to zero. The rate of increase did not depend on age (interaction OR 0.97 per 10 years of age and doubling of attempt number, 95% CI [0.90, 1.05]; $p = 0.484$).

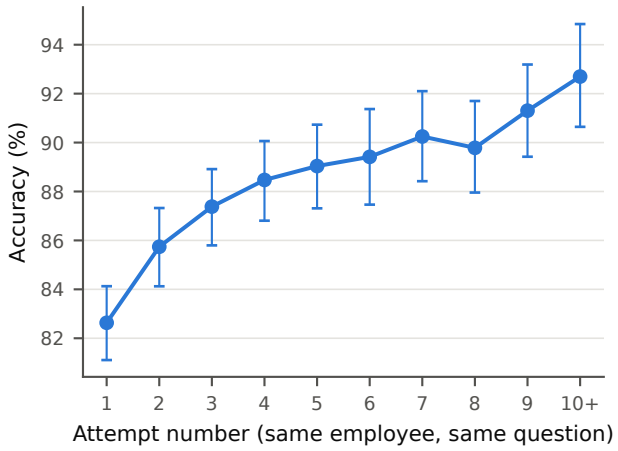


Figure 1: Accuracy by attempt number in all sequences available at a given attempt (their number decreases with attempt number). Error bars: 95% CI (bootstrap). Attempts 10 and later were pooled. The course in the cohort is shown in Figure 3.

Table 4: Accuracy at successive attempts in the cohort, overall and by employee age group.

Age group (years)	Overall	20–29	30–39	40–49	50–59	60–69	Missing
Employees	124	9	15	35	39	13	13
Sequences, n	4890	199	664	1784	1689	471	83
Attempt 1 (%)	81.9	81.4	77.0	81.6	83.8	82.4	86.7
Attempt 2 (%)	85.6	80.4	79.4	85.7	88.0	86.6	94.0
Attempt 3 (%)	87.2	85.4	81.5	88.5	88.3	86.0	95.2
Attempt 4 (%)	88.4	87.9	82.7	89.5	89.7	86.2	96.4
Attempt 5 (%)	89.0	86.9	84.2	90.4	90.1	86.4	97.6
Change, attempt 5 vs 1 (pp)	7.2	5.5	7.2	8.8	6.3	4.0	10.8
95% CI	[5.5, 8.9]	[0.0, 14.7]	[3.2, 13.5]	[5.9, 11.6]	[4.3, 8.1]	[0.0, 9.3]	[1.9, 19.8]

Cohort: sequences of at least 5 attempts, attempts 1 to 5. pp: percentage points; 95% CIs of the change were obtained with the bootstrap. The association between the rate of improvement and age was estimated by the logistic regression described in Section 2.3 (23,140 responses from employees of known age).

3.3 Outcome of the previous attempt and interval (RQ2)

The increase in accuracy results from two opposing processes: error correction and forgetting. The schedule did not bias their comparison, because a question returned after a correct and after an

incorrect response with similar probability (73.5% and 74.5%) and after a similar interval (median 37 and 39 days). After a correct response, the next one was correct in 92.6% [91.8, 93.4] of cases, and forgetting occurred in 7.4%. After an incorrect response, an error correction followed in 58.3% [55.5, 61.2] of cases. These rates refer to different groups of attempts, so the numbers of events were also compared: there were 3382 error corrections and 2719 forgetting events. The predominance of corrections explains the increase in accuracy. At constant error-correction and forgetting rates, accuracy would tend toward an equilibrium level of 88.8%. Accuracy in the cohort at attempt 5 (89.0%) was close to this level, but at attempts 10 and later it was 92.7% and exceeded it. This is because the forgetting rate was not constant but depended on the interval and on the number of consecutive correct responses.

The forgetting rate increased with the interval, from 4.8% at intervals shorter than 14 days to 11.5% at intervals of at least 240 days (Figure 2). The error-correction rate changed less with the interval (60.9% and 58.1% in the same bins).

The forgetting rate also depended on the number of consecutive correct responses. After one correct response, it was 11.9% [10.7, 13.0], after two 7.6%, after three 5.0%, and after at least 4 3.2% [2.7, 3.7] (Table 5). The difference persisted at every interval: at intervals shorter than 14 days, the forgetting rate was 7.7% after one and 2.3% after at least three correct responses, and at intervals of at least 240 days 13.9% and 5.2%, respectively. Error corrections were the least durable. After a correction, an error relapse occurred in 24.9% [22.9, 26.8] of cases ($n = 2457$), and in 73.6% [69.5, 77.7] of these the employee chose the same distractor as before the correction. This share was similar to the percentage of repeated choices of the same distractor after two successive errors (71.8%; Section 3.4).

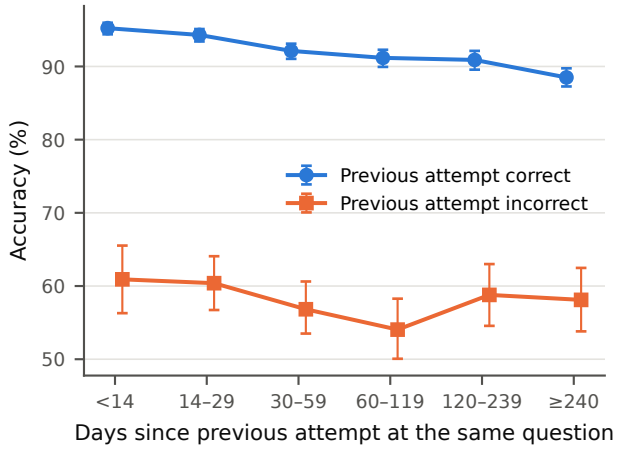


Figure 2: Accuracy by interval, separately for attempts preceded by a correct response (circles) and by an incorrect response (squares). For attempts preceded by a correct response, the forgetting rate is 100% minus accuracy; for attempts preceded by an incorrect response, accuracy is the error-correction rate. Error bars: 95% CI (bootstrap).

Table 5: Forgetting rate by the number of consecutive correct responses given immediately before a given attempt.

Consecutive correct before attempt	1	2	3	≥4
Attempts, n	12,318	8276	5597	10,734
Employees	159	147	137	123
Forgetting rate (%)	11.9	7.6	5.0	3.2
95% CI	[10.7, 13.0]	[6.7, 8.5]	[4.2, 5.8]	[2.7, 3.7]

Attempts preceded by a correct response. Forgetting rate: percentage of incorrect responses. 95% CIs were obtained with the bootstrap.

3.4 Persistent errors (RQ3)

Most errors were corrected: in 81.8% of the 2068 sequences that began with an incorrect response and comprised at least two attempts, the employee later answered correctly at least once. The chance of correction, however, decreased with each successive error. After one error, accuracy was 64.4% [62.2, 66.8], after two consecutive errors 48.1%, after three 41.8%, and after at least 4 34.2% [27.6, 40.9] (Table 6). Part of this decrease may be due to several consecutive errors occurring mainly in more difficult questions.

Repeated errors were not consistent with random choice. After two successive errors, the employee chose the same distractor in 71.8% [69.4, 74.1] of cases ($n = 2420$), whereas 47.2% would be expected under random choice and 60.0% under choice proportional to distractor popularity. This percentage did not depend on the interval (70.6% at intervals shorter than 30 days, 73.2% at 30–89 days and 71.6% at intervals of at least 90 days). Repeated errors were concentrated in few questions: 14 of the 145 questions accounted for 39.6% of them.

Table 6: Accuracy by the number of consecutive errors made immediately before a given attempt.

Consecutive errors before attempt	0	1	2	3	≥4
Attempts, n	36,925	4024	1074	409	295
Employees	159	149	110	80	48
Accuracy (%)	92.6	64.4	48.1	41.8	34.2
95% CI	[91.9, 93.4]	[62.2, 66.8]	[44.7, 51.9]	[36.4, 47.3]	[27.6, 40.9]

Attempts from the second onward. Column 0: attempts preceded by a correct response. 95% CIs were obtained with the bootstrap.

3.5 Summary of results

Figure 3 summarizes the results in the cohort. Accuracy increases from attempt to attempt (RQ1), error corrections outnumber forgetting events (RQ2), and at every attempt from the second onward, part of the incorrect responses are repeated errors (RQ3).

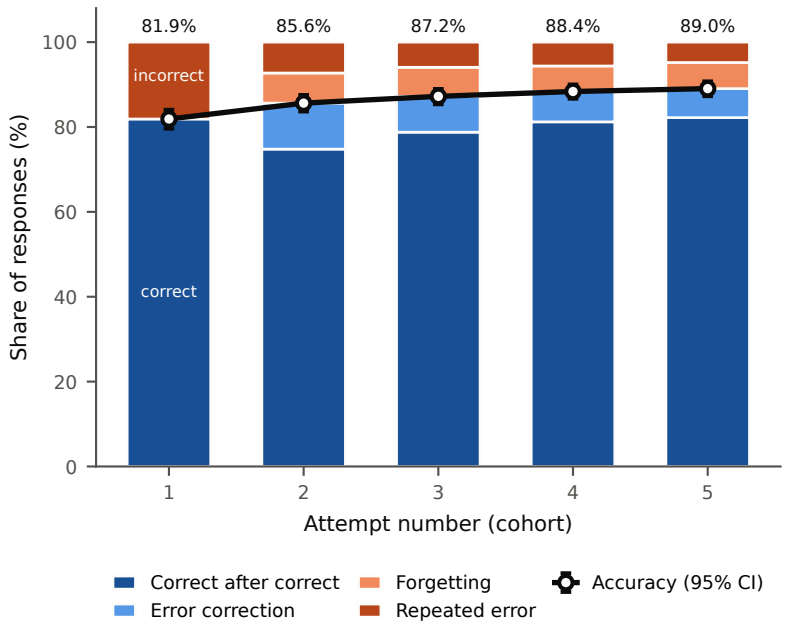


Figure 3: Accuracy and response state in the cohort (4890 sequences, attempts 1 to 5). Bars: share of responses by the outcome of the current and the previous attempt (Section 2.3); at attempt 1, only correct and incorrect responses. Line: accuracy with 95% CI (bootstrap), values above the bars.

4 Discussion

In the bank's microlearning program, response accuracy increased with each return of a question, also in the cohort in which the same sequences were compared. The increase resulted from error corrections outnumbering forgetting events. Some errors, however, persisted despite feedback and consisted in choosing the same distractor again. The direction of change is consistent with studies of spaced education in medical training. In urology residents, the knowledge advantage gained from periodically recurring questions persisted two years after the end of the program [7], and a systematic review of studies on clinicians' continuing professional development indicated a beneficial effect of this method on the acquisition and retention of knowledge [10].

The conditions of those studies, however, differed from those of the present analysis. The participants were health professionals, and the effect was usually assessed with a test administered outside the program, in comparison with a control group. In the program studied here, the participants were bank employees, and the increase in accuracy was observed only in responses given within the program itself. The agreement therefore concerns the direction of change, not the size of the effect or its cause. The results do, however, show what the increase in accuracy consisted of, which a final test score does not reveal. It was the difference between two opposing processes, error correction and forgetting (3382 vs. 2719 events), and the same overall increase could also arise with more frequent forgetting if it were accompanied by a correspondingly larger number of corrections. Aggregate measures also do not detect errors that persist despite feedback.

Forgetting was associated with three factors. The first was the passage of time: forgetting was more frequent at longer intervals, in line with research on the spacing between repetitions [4]. The second was a short history of correct responses. Knowledge confirmed by one correct response was

forgotten two to four times more often than knowledge confirmed several times in a row, at every interval. Similarly, in studies of repeated practice, each additional successful retrieval increased the durability of knowledge [11]. The third factor was error relapse. Error correction proved the least durable, and when it did not hold, the employee usually returned to the distractor chosen before the correction. The return of original errors after feedback has also been described in laboratory studies [3]. The correct answer may thus not have replaced the misconception but coexisted with it.

Repeated errors differed in character from forgetting. Employees chose the same distractor more often than chance would predict, regardless of the interval, and each successive error reduced the chance of correction. Such a pattern is consistent with a persistent misconception rather than with a lack of knowledge, which would lead to guessing. A selected distractor can become fixed in memory, and feedback reduces this effect [2]. In the program studied, the feedback gave the correct answer and an explanation but did not address the selected distractor, although explaining why the selected answer is wrong is usually more effective [15]. The concentration of repeated errors in 14 questions further suggests that some of them resulted from question design, e.g., an ambiguous distractor or content inconsistent with the bank's practice. Such questions can be identified by standard distractor analysis [6].

In one-off training, e.g., in information security, the effects faded after several months [12]. The present results indicate that knowledge requires further repetition also in a program with periodically recurring questions, and that the way these repetitions are scheduled can be improved. Because the interval did not depend on the outcome, mastered questions returned as often as questions in which the employee had only just corrected an error. An adaptive schedule could lengthen the interval after several consecutive correct responses and shorten it after an error correction, when the risk of relapse is highest. Personalized review improved long-term retention compared with a fixed schedule [8], and models of forgetting rate based on response history are already used in language learning applications [14]. A second change could be feedback addressing the selected distractor, introduced first in the questions with the largest number of repeated errors.

4.1 Limitations and future research

The study was observational and had no control group, so the increase in accuracy cannot be attributed solely to the program, as employees also acquired knowledge in their everyday work. Because the questions were repeated, the data do not allow consolidation of knowledge to be distinguished from memorization of the answer to a specific question. Response time and employee confidence were not recorded, so forgetting cannot be separated from inattention or from a lucky guess at an earlier attempt, which learner models capture as slip and guess [5]. Such a source of part of the forgetting is suggested by its rate of 4.8% already at intervals shorter than 14 days. The longest intervals may have included periods when the program was suspended, so changes in question content may have inflated the forgetting rate at these intervals. The comparisons were descriptive and were not corrected for multiple testing. The confidence intervals accounted for the dependence between responses of the same employee, but not of the same question. The associations with the number of consecutive correct responses or errors may therefore partly reflect differences in difficulty between questions.

Future studies could compare, in a randomized design, a fixed with an adaptive schedule and standard feedback with feedback addressing the selected distractor, using equivalent versions of questions that would separate knowledge from memorization of the answer. Recording response confidence would make it possible to test whether persistent errors concern high-confidence beliefs [3]. Microlearning outcomes should also be linked to the application of knowledge at work, e.g., to the number of procedural errors [1].

5 Conclusions

In a program with periodically recurring questions, the response accuracy of bank employees increased from attempt to attempt. Knowledge nevertheless required further repetition, because it was forgotten over time, particularly after the first correct response and after an error correction. Some errors persisted despite feedback and consisted in choosing the same distractor again. A schedule adapted to employee performance and feedback addressing the selected distractor may increase the effectiveness of such programs, but their effect requires confirmation in studies with a control group.

Declarations

Ethics. The study used fully anonymized data provided by the bank with its written consent of 3 August 2026. Employee coding and data protection rules are described in Section 2.2.
Data availability. The data used in the study are the property of the bank and are not publicly available.
Funding. The study received no funding.
Competing interests. M. Szerszeń is an employee of the bank in which the study was conducted. The other authors declare no competing interests.
Use of AI tools. During the preparation of this article, the authors used Anthropic Claude Sonnet 5.5 for (1) language editing of the description of the results of the analysis performed by the authors and (2) translation of the text from Polish into English. Data analysis, interpretation of the results and conclusions were carried out by the author. The authors verified and edited the entire text and take full responsibility for its content.

References

[1] Brian D. Blume, J. Kevin Ford, Timothy T. Baldwin, and Jason L. Huang. Transfer of training: a meta-analytic review. *Journal of Management*, 36(4):1065–1105, 2010. doi:10.1177/0149206309352880.

[2] Andrew C. Butler and Henry L. Roediger. Feedback enhances the positive effects and reduces the negative effects of multiple-choice testing. *Memory & Cognition*, 36(3):604–616, 2008. doi:10.3758/MC.36.3.604.

[3] Andrew C. Butler, Lisa K. Fazio, and Elizabeth J. Marsh. The hypercorrection effect persists over a week, but high-confidence errors return. *Psychonomic Bulletin & Review*, 18(6):1238–1244, 2011. doi:10.3758/s13423-011-0173-y.

[4] Nicholas J. Cepeda, Harold Pashler, Edward Vul, John T. Wixted, and Doug Rohrer. Distributed practice in verbal recall tasks: a review and quantitative synthesis. *Psychological Bulletin*, 132(3):354–380, 2006. doi:10.1037/0033-2909.132.3.354.

[5] Albert T. Corbett and John R. Anderson. Knowledge tracing: modeling the acquisition of procedural knowledge. *User Modeling and User-Adapted Interaction*, 4:253–278, 1994. doi:10.1007/BF01099821.

[6] Mark J. Gierl, Okan Bulut, Qi Guo, and Xinxin Zhang. Developing, analyzing, and using distractors for multiple-choice tests in education: a comprehensive review. *Review of Educational Research*, 87(6):1082–1116, 2017. doi:10.3102/0034654317726529.

[7] B. Price Kerfoot. Learning benefits of on-line spaced education persist for 2 years. *Journal of Urology*, 181(6):2671–2673, 2009. doi:10.1016/j.juro.2009.02.024.

[8] Robert V. Lindsey, Jeffery D. Shroyer, Harold Pashler, and Michael C. Mozer. Improving students' long-term knowledge retention through personalized review. *Psychological Science*, 25 (3):639–647, 2014. doi:10.1177/0956797613504302.

[9] R. Charles Murray, Steven Ritter, Tristan Nixon, Ryan Schwiebert, Robert G. M. Hausmann, Brendon Towle, Stephen E. Fancsali, and Annalies Vuong. Revealing the learning in learning curves. In *Artificial Intelligence in Education (AIED 2013)*, volume 7926 of *LNCS*, pages 473–482, 2013. doi:10.1007/978-3-642-39112-5_48.

[10] Jane L. Phillips, Nicole Heneka, Priyanka Bhattarai, Clare Fraser, and Tim Shaw. Effectiveness of the spaced education pedagogy for clinicians' continuing professional development: a systematic review. *Medical Education*, 53(9):886–902, 2019. doi:10.1111/medu.13895.

[11] Katherine A. Rawson, John Dunlosky, and Sharon M. Sciartelli. The power of successive relearning: improving performance on course exams and long-term retention. *Educational Psychology Review*, 25(4):523–548, 2013. doi:10.1007/s10648-013-9240-4.

[12] Benjamin Reinheimer, Lina Aldag, Peter Mayer, Mattia Mossano, Reyhan Duezguen, Bettina Lofthouse, Tatiana von Landesberger, and Melanie Volkamer. An investigation of phishing awareness and education over time: when and how to best remind users. In *Sixteenth Symposium on Usable Privacy and Security (SOUPS 2020)*. USENIX Association, 2020.

[13] Henry L. Roediger and Jeffrey D. Karpicke. Test-enhanced learning: taking memory tests improves long-term retention. *Psychological Science*, 17(3):249–255, 2006. doi:10.1111/j.1467-9280.2006.01693.x.

[14] Burr Settles and Brendan Meeder. A trainable spaced repetition model for language learning. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, pages 1848–1858, 2016. URL `https://aclanthology.org/P16-1174`.

[15] Valerie J. Shute. Focus on formative feedback. *Review of Educational Research*, 78(1):153–189, 2008. doi:10.3102/0034654307313795.

[16] Allison D. Taylor and Woei Hung. The effects of microlearning: a scoping review. *Educational Technology Research and Development*, 70:363–395, 2022. doi:10.1007/s11423-022-10084-1.